

Image Classification using Deep Learning on the CIFAR-10 Dataset

Project Documentation

1. Dataset Description

This project uses the **CIFAR-10** dataset, one of the most widely used benchmark datasets for image classification. The objective is to classify images into one of ten predefined categories using deep learning techniques. The dataset consists of **60,000 RGB images** of size **32 × 32 pixels**, where **50,000 images** are used for training and **10,000 images** are used for testing.

Classes

- Airplane
- Automobile
- Bird
- Cat
- Deer
- Dog
- Frog
- Horse
- Ship
- Truck

The dataset was loaded using **TensorFlow/Keras** and explored to understand its structure, image dimensions, and class distribution.

2. Data Preprocessing

Image Augmentation

To improve the model's ability to generalize and reduce overfitting, the following augmentation techniques were applied:

- Random Horizontal Flip
- Random Rotation

- Random Zoom

Normalization

Pixel values were converted to floating-point values and normalized by dividing by 255, scaling them to the range **0–1**.

Data Splitting

The dataset was divided into:

- Training Set
- Validation Set
- Test Set

The validation set was used to monitor the model's performance during training and implement **Early Stopping**.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed before model development to understand the dataset.

Analysis Performed

- Displayed sample images from each class.
- Verified image dimensions.
- Checked the total number of images.
- Examined class distribution.
- Confirmed that the dataset was balanced.

Observations

- Each class contains approximately the same number of images.
 - Images have a low resolution of **32 × 32 pixels**.
 - Several object categories share similar visual characteristics, making classification challenging.
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4. Model Development

Custom CNN

A custom Convolutional Neural Network (CNN) was developed consisting of:

- Data Augmentation Layer
- Conv2D Layer (64 Filters)
- MaxPooling Layer
- Conv2D Layer (128 Filters)
- MaxPooling Layer
- Conv2D Layer (256 Filters)
- Flatten Layer
- Dense Layer (256 Neurons)
- Dropout Layer (0.3)
- Softmax Output Layer

Model Training

The CNN was trained using:

- Adam Optimizer
- Sparse Categorical Crossentropy Loss
- Early Stopping
- Batch Size: 64
- Maximum Epochs: 30

Model Evaluation

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

5. Model Optimization

Hyperparameter Tuning

Hyperparameter tuning was performed using **Keras Tuner (Random Search)**.

The following parameters were optimized:

- Number of Convolution Filters
- Dense Layer Units
- Dropout Rate
- Learning Rate

Best Hyperparameters

- Conv1 Filters: **64**
- Conv2 Filters: **128**
- Conv3 Filters: **256**
- Dense Units: **256**
- Dropout Rate: **0.3**
- Learning Rate: **0.0005**

After optimization, the CNN achieved a **test accuracy of 77%**, outperforming the baseline CNN.

Transfer Learning

Transfer learning was implemented using the pretrained **MobileNetV3Small** model with **ImageNet** weights.

The implementation included:

- Image Resizing
- Frozen Pretrained Layers
- Custom Classification Layer
- Fine-Tuning
- Early Stopping

The transfer learning model achieved a **test accuracy of 71%**. Although it performed well, it did not surpass the hyperparameter-tuned custom CNN due to the small image size of the CIFAR-10 dataset.

6. Model Performance Comparison

Model	Test Accuracy
Baseline CNN	72%
Hyperparameter Tuned CNN	77%
MobileNetV3Small (Transfer Learning)	71%

The hyperparameter-tuned CNN produced the best overall performance.

7. Results and Discussion

The confusion matrices and classification reports showed that the models correctly classified most images, particularly **automobiles, ships, trucks, frogs, and airplanes**. However, visually similar categories such as **cats and dogs, birds and frogs**, and **deer and horses** remained challenging due to the low image resolution.

Hyperparameter tuning improved the CNN's learning capability and produced the highest classification accuracy. Although transfer learning reduced training time by utilizing pretrained ImageNet features, the MobileNetV3Small model achieved lower accuracy than the tuned CNN on this dataset.

8. Tools and Technologies

Programming Language

- Python

Libraries

- TensorFlow
- Keras
- Keras Tuner
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

9. Project Structure

The project contains:

- Dataset Loading
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Data Augmentation
- Custom CNN Model
- Hyperparameter Tuning
- Transfer Learning (MobileNetV3Small)
- Model Evaluation
- Performance Comparison
- Model Saving

10. Conclusion

This project successfully developed an image classification system using the **CIFAR-10** dataset by implementing multiple deep learning approaches. The workflow included data preprocessing, data augmentation, custom CNN development, hyperparameter tuning, transfer learning with MobileNetV3Small, and comprehensive model evaluation using accuracy, precision, recall, F1-score, and confusion matrices. Among all the implemented models, the **hyperparameter-tuned CNN achieved the best performance with a test accuracy of 77%**, outperforming both the baseline CNN (72%) and the transfer learning model (71%). The results demonstrate that careful hyperparameter optimization significantly improves classification performance, while transfer learning provides an efficient alternative but may not always outperform a well-designed custom CNN on small, low-resolution datasets such as CIFAR-10. Overall, the project successfully demonstrates the complete deep learning workflow for image classification, from data preprocessing to model optimization and performance evaluation.